EVERY FINITE CONSTRUCTIBLE STRATIFICATION OF AN AFFINE SCHEME IS REFINED BY THE STRATIFICATION ATTACHED TO A FINITE SET OF ELEMENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the presentation of the constructible strata through quasi-compact opens and their complements, the replacement of those opens by finitely many basic opens, and the verification that the vanishing-pattern stratification refines the given one.

Editorial note. The label lemma-refine names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-refine –start 770 –end 787 –local-id 28006 returned RESULT=OK (L28006 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfB.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Presentation of a constructible subset as a finite union of differences of quasi-compact opens. Topology, the description of constructible subsets of a spectral space as finite unions of locally closed subsets built from quasi-compact opens. Supplies the writing of each stratum T_i as a finite union of intersections U_{i,j} cap (V_{i,j})^c.
- Description of quasi-compact opens by basic opens. Algebra and Schemes, the description of a quasi-compact open of the spectrum of a ring as a finite union of basic opens D(f). Supplies the elements f_{i,j,k} and g_{i,j,l} whose basic opens cut out the U_{i,j} and V_{i,j}.
- The stratification attached to a finite set of elements. Same chapter, the definition of the stratification whose strata are labelled by the vanishing pattern of the elements of a finite set. Supplies the stratification that the construction refines the given one.

Source reference. Original equation 096X: sources/stacks-proetale/pinned-original-source.tex, source lines 765–765. Prerequisite pointer invoked at proof line(s) 774; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:proetale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $X = \mathrm{Spec}(A)$ as above. Given any finite stratification $X = \coprod T_i$ by constructible subsets, there exists a finite subset $E \subset A$ such that the stratification (096X) refines $X = \coprod T_i$.*

Proof. We may write $T_i = \bigcup_j U_{i,j} \cap V_{i,j}^c$ as a finite union for some $U_{i,j}$ and $V_{i,j}$ quasi-compact open in X. Then we may write $U_{i,j} = \bigcup D(f_{i,j,k})$ and $V_{i,j} = \bigcup D(g_{i,j,l})$. Then we set $E = \{f_{i,j,k}\} \cup \{g_{i,j,l}\}$. This does the job, because the stratification (096X) is the one whose strata are labeled by the vanishing pattern of the elements of E which clearly refines the given stratification. $\square$